Comparison of Tree Encoding Schemes for Biobjective Minimum Spanning Tree Problem

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Abstract—**Minimum Spanning Trees (MST) problem is a classical problem in operation research and network design problem is an important application of it. Minimum Spanning Tree (MST) problem can be solved efficiently, but its Biobjective versions are NP hard. In this paper, we compare three tree encoding schemes using Biobjective evolutionary algorithm. Three different tree encoding methods in the evolutionary algorithms are being used to solve three different instances of Biobjective Minimum Spanning Tree problem; comparative study of the tree encoding schemes used is done on the basis of Pareto optimal front obtained. Our approach involves Biobjective Minimum Spanning Tree problem using Nondominated Sorting Genetic Algorithm II (NSGAII). We compare Edge Set encoding, Prüfer encoding, Characteristic Vector encoding using evolutionary algorithm for Biobjective Minimum Spanning Tree , we find that edge sets encoding performs better than Prüfer and Characteristic Vector for Biobjective Minimum Spanning Tree problem while we are solving Biobjective Minimum Spanning Tree problem using Nondominated Sorting Genetic Algorithm II (NSGAII).**

Keywords- Minimum Spanning Tree; Nondominated Sorting Genetic Algorithm; Biobjective optimization scenario; Evolutionary Algorithm.

I. INTRODUCTION

The Minimum Spanning Tree (MST) is classical combinatorial optimization problem. An important class of this combinatorial optimization problem is network design. This class is used for modeling of a number of practical systems, such as Telecommunication, Gas, and Computer Network. In all these cases, the use of optimization is important because the improved implemented system leads to considerable financial saving and better performance.

Suppose G is undirected, weighted, connected graph, the minimum spanning tree problem is to find a tree with minimum total cost. It is well studied in past and many efficient polynomial time algorithms have been developed by Prim[1], Kruskal[2], Dijakstra[3], out of these one of the classic and favored algorithm is Prim's algorithm[1].

Definition of Biobjective Minimum Spanning Tree is not an extension from single objective Minimum Spanning Tree to Biobjective. In general, due to the conflicts between the two objectives the optimal solution of problem cannot be found. Pareto optimal solution [6] set is the best way to get real solution of Biobjective problem. Further it is not an easy task because of the NP hard nature of problem even in unconstrained form. The Biobjective Minimum Spanning Tree is commonly found in network design oriented application. EA can be used to solve Network design problems and other NP hard problems, as it has emerged as an appropriate method to solve these variants of Minimum Spanning Tree. It is important to note that the choice of representation scheme in EA is critical aspect of algorithm designing particularly in network design optimization [7].Use of inadequate representation can lead noticeable reduction in Algorithm performance suffers.

In literature, Zohu and Gen[5] proposed a Prüfer encoded multiobjective EA using Non Dominated Sorting method [16] is the only one EA purposed for mc-MST, to our knowledge. In this paper Zohu and Gen have applied their method to randomly generated Biobjective mc- MSTs .The basic goal is the comparison of their EA with the enumeration method results. Later, It was found by Knowles and Corne [19] that the enumeration algorithm was not correct. Then Knowles and Corne [19] presented AESSEA (Archived Elitist Steady State Evolutionary Algorithm) algorithm in two versions: AESSEA + Prüfer and AESSEA+ Direct/RPM. In first version the Prüfer method for encoding the solutions and the operators suggested by of Zhou and Gen[5] are used. In second version uses Raidl's[12] method of encoding and operators suggested by him for degree constrained minimum spanning tree problem.

Three evolutionary algorithms for Biobjective minimum spanning tree has been described in this paper. First one follows NSGA II and Edge set based encoding and Raidl [12] operators for direct encoding. Second one employs NSGA II and Prüfer based encoding described in [5] and standard operators for EA. Third one uses NSGA II and Characteristic Vector, the performance of all the EAs is compared on a test suite of Biobjective minimum spanning tree problems.

Rest of the paper is organized as follows: Biobjective minimum spanning tree problem is described in Section II. Section III defines the properties desired for representation of tree in evolutionary algorithm (EA) and three encoding schemes (Edge set, Prüfer, Characteristic Vector) and their operators. Section IV discusses multiobjective evolutionary algorithm used. The numerical experiments and results have been discussed in Section V. Finally conclusion follows in Section VI.

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II. BIOBJECTIVE MINIMUM SPANNING TREE PROBLEM

Suppose G = (V,E) is an undirected, connected , weighted complete graph then a spanning tree for this graph is a sub graph G_T =(V,E_T) where $E_T \in$ E and also contains all vertices in V, and also connects all vertices with exactly |V|-1 edges ensuring that there are no cycles.

The minimum spanning tree problem can be defined as to find a tree T whose total cost

$$WT = \sum_{(u,v)\in ET} w(u,v) \qquad (1)$$

is minimum. The diameter of a tree can be defined as largest number of edges on any path on tree T. Problem of finding a MST that does not contain a path having more than k edges is called DCMST problem, for all 4≤ k ≤ (n-1) where n=|V| the DCMST problem is known to be NP hard [8]. Biobjective spanning tree problem can be defined by considering diameter as an objective along with edge cost.

III. DESIARABLE PROPERTIES OF TREE REPRESENTATION AND TREE REPRESENTATION SCHEMES

A. Properties desired for tree representation in EA

There are many factors that influence the performance of EA; one of the most important factors is representation or encoding the solution for tree representation. Reference [9] states the following properties:

- It should be capable of representing all possible trees.
- The encoding should be unbiased in the sense that all trees are equally represented; i.e., the same number of encodings should represent all trees.
- The encoding should represent all feasible trees, and genetic operators result should always be feasible trees.
- It should be inexpensive in evaluating the fitness function and constraints while going back and forth between the encoded tree and its representation.
- The encoding should possess locality, i.e. small changes in the representation make small change in the tree.

Three tree representations will be discussed in the following sections and evaluate them on the basis of the properties described.

B. Representation for spanning tree

1) Prüfer Numbers: Cayley gave one of the first theorem in graphical enumeration which states that there are n^{n-2} distinct spanning tree in n vertices complete graph.Later, this theorem was proven by Prüfer [11] and he established a one-to-one correspondence between spanning trees and a string of length n-2 [10,11] over an integer between 1 to n. This string is denoted as Prüfer number, and the mapping of genotype-phenotype is called Prüfer number encoding. A unique tree can be derived with n nodes from the Prüfer number of length (n – 2) and vice versa.

Encoding details are discussed in Reference [5], and are well known. In this encoding scheme, a chromosome size is a string of (n - 2) digits and each digit can vary between 1 and n. In our implementation the encoding uses uniform crossover and single gene mutation for Prüfer encoding.

2) The edge set encoding: The Edge Sets encoding proposed by Raidl [12, 13] is a direct way to represent a spanning tree as a set of edges. That is why it is called the Edge set representation. Special mutation and crossover operators are used by Raidl for this representation. In Raidl's mutation, a new edge replaces the existing edges of tree. In operations, it works as follows. Firstly, an edge currently not in the tree is chosen to insert, and choice is made to favor lower cost edges. Naturally, it will create a cycle. Then a random choice is made among the edges in that cycle excluding the newly inserted edge, and chosen edge is removed.

Raidl's crossover operator principle is to inherit as many edges as possible from the parent trees. In first step, a child is initialized to have all edges which are common in both parent solutions and new solution will always include these edges. In the second step, all those edges which are in either parent but not in both are checked for inclusion, and included if its inclusion is feasible. If newly constructed child is not a spanning tree at the end of the process, randomly chosen edges between its components are included until it is not a spanning tree.

3) Characteristic Vector: A binary string whose position corresponds to item in a set is called Characteristic vector and each bit of that bit string indicates whether or not the corresponding item is included in structure represented by string. When a spanning tree is represented by such a binary string (Characteristic Vector), the edges of graph are items and Characteristic Vector indicates whether the each edge is included or not in the tree. Formally, A graph of n nodes consists of $n(n-1)/2$ possible links, so a Characteristic Vector of length $l= n(n-1)/2$ is needed to encode n node graph. Each and every link must be numbered and they must be assigned a position in a Characteristic Vector.

To encode graphs, Characteristic Vector is a common approach and is used to by several researchers to represent spanning tree in EAs. Some of them are Davis et al. [17] and Piggott and Suraweera [18]. The EAs used by these researcher applied usual crossover and mutation operator.

IV. ALGORITHM

A. Nondominated Sorting Genetic Algorithm II (NSGA II)

NSGA II is an improved version of the earlier algorithm NSGA [15]. NSGA II is computationally more efficient, elitist and uses a crowded comparison operator to keep diversity and does not use any additional parameters for keeping diversity. Reference [14] states the following Algorithm:

$R_t = P_t \cup Q_t$
F=fast-non-dominated-sort(R_t)
$P_{t+1} = \emptyset$ and i=1
Until $P_{t+1} + F_i \leq N$
Crowding-distance-assignment(F_i)
$P_{t+1} = P_{t+1} \cup F_i$
i=i+1
Sort (F_i , $\prec_n$)
$P_{t+1} = P_{t+1} \cup F_i$ [1: (N- P_{t+1})]
Q_{t+1}=make-new-pop(P_{t+1})
t=t+1

For each generation t, an external population of N parent individuals Pt is maintained by the algorithm and a child population Qt is created from the parents. Two different global ranking measures are used to sort both populations lexicographically and then both the population are combined together. The first measure is the non-dominated sorting, which is the result of fast nondominated sort procedure. For details of its implementation, refer [3]. The outcome of this procedure is the assignment of a rank to each solution in the set Rt. If the two solutions have same rank then they do not dominate each other, but there exists at least one dominating individual of lower rank for each solution of rank r> 1. All solutions that are in the Pareto set have rank 1. Thus, a total ordering of the set of solutions in the algorithm for each generation is implied by the rank value. For more competitive ordering a secondary ranking measure to each solution is also assigned by NSGA II. Second measure is crowding distance, and is well suited for later stage of the algorithms' application. When population is already close to the true Pareto front of the problem, solution s are forced to keep distance to their neighboring solutions in objective space by crowding distance. For more detail refer Deb et al. [14]).

V. EXPERIMENT AND RESULTS

Three instances of complete graphs of size 15, 25, 50 vertices having uniformly and randomly distributed weighted edges have been tested using NSGA-II. There are two objectives for all the three instances; first one is to minimize the cost of constructed spanning tree and other one is to minimize the diameter of it.

To solve problem instances using Biobjective EA, the parameters of Biobjective EA are set as follows: crossover probability:0.80, mutation probability: 0.05, no. of generations: 2500.

All Pareto optimal solutions obtained for various Biobjective Minimum Spanning Tree (MST) problem instances using NSGA II + Edge set, NSGA II + Prüfer and NSGA II + Characteristic Vector encoding schemes are shown in Fig. 1, Fig. 2, Fig. 3 after 2500 run.

The performance of NSGA II + Prüfer method is worst and finds no true Pareto optima. With larger problems obtained Pareto front of NSGA- II + Prüfer is smaller than the other two. The NSGA II + Characteristic Vector method performs better than the NSGA- II + Prüfer and converges towards the true Pareto optima as the size of the problem is larger due to high locality and heritability of Characteristic Vector encoding. But the NSGA II + Edge set outperforms NSGA- II + Prüfer and NSGA II + Characteristic Vector both. These results are illustrated in Fig. 1, Fig. 2, Fig. 3, and clearly indicate NSGA II + Edge set approaches closely to the true Pareto front in comparison to NSGA II + Prüfer and NSGA II + Characteristic Vector. As problem size increases the difference between the NSGA II + Edge set and NSGA II + Prüfer increases, while the difference between the NSGA II + Edge set and NSGA II + Characteristic Vector decreases due to high locality and heritability of Characteristic Vector encoding.

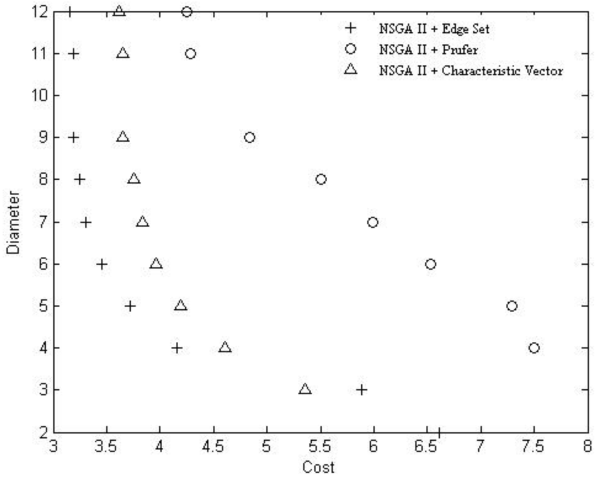

Figure 1. Combined Pareto optimal fronts discovered by NSGA II + Edge set and NSGA II + Prüfer + NSGA II + Characteristic Vector for 15 Vertices graph.

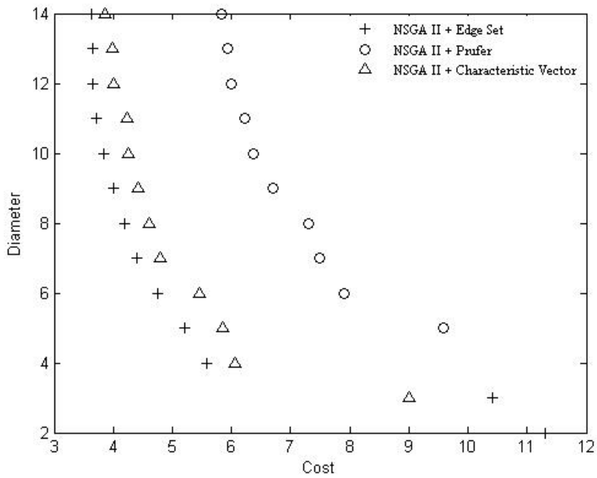

Figure 2. Combined Pareto optimal fronts discovered by NSGA II + Edge set and NSGA II + Prüfer + NSGA II + Characteristic Vector for 25 Vertices graph.

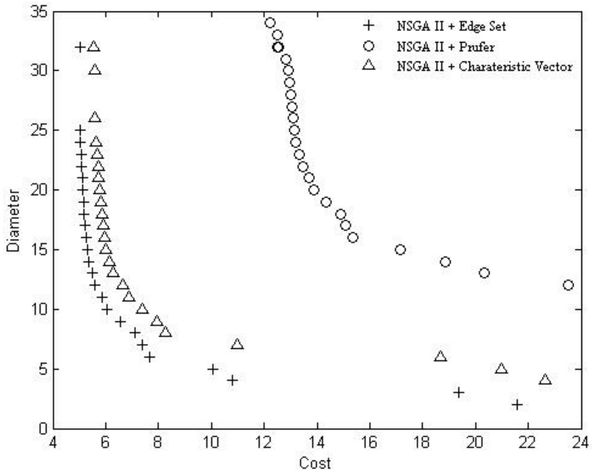

Figure 3. Combined Pareto optimal fronts discovered by NSGA II + Edge set and NSGA II + Prüfer + NSGA II + Characteristic Vector for 50 Vertices graph.

VI.CONCLUSION

Comparison of three classical tree encoding schemes using Biobjective Evolutionary algorithm has been presented in this paper. The comparison has been made on the basis of their performance on a NP hard problem (i.e. Biobjective Minimum Spanning Tree (MST) problem). The experimental results obtained show the dominance of Edge set encoding scheme over Prüfer and Characteristic Vector encoding schemes for Biobjective Minimum Spanning Tree (MST) problem. Other proposed tree encoding schemes such as Dandelion code, Blob code, Happy code, Network random key encoding have not been addressed in our experiment. So, the work can be extended to compare these proposed encoding schemes using Biobjective Evolutionary Algorithm.

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